



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment-3.2

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Semester: 5th

Subject Name: DAA

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Section/Group: 23KRG-2B

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Subject Code: 23CSH-301

1. **Aim:** Develop a program and analyze complexity to find all occurrences of a pattern P in a given string S.
2. **Objective:** Analyze to find all occurrences of a pattern P in a given string S
3. **Input/Apparatus Used:** String is taken as input in order to find pattern from it.

4. Procedure/Algorithm: Pseudocode:

1. We will first create the LPS array.
 2. Initialize two variables - „strIdx“ and „patIdx“ to iterate over the string and the pattern, respectively.
 3. If „pat[patIdx]“ equals „str[strIdx]“, we will increment both the indexes.
 4. When „patIdx“ equals the length of the pattern, this means that the pattern is found in the string. Therefore we print the index and set „patIdx“ = LPS[patIdx-1].
 5. If „pat[patIdx]“ is not equal to „str[strIdx]“, we update the patIdx with the last index that matches with „str[strIdx]“ using the LPS array.
- Doing this, we will find all occurrences of the pattern in the string.

5. Code:

```
Code.cpp X
Code.cpp > ...
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  int main() {
6      string text, pattern;
7      cout << "Enter the text: ";
8      getline(cin, text);
9      cout << "Enter the pattern: ";
10     getline(cin, pattern);
11
12     int n = text.size();
13     int m = pattern.size();
14     int d = 256;
15     int q = 101;
16     int h = 1;
17     int p = 0, t = 0;
18
19     for(int i=0;i<m-1;i++)
20         h = (h * d) % q;
21
22     for(int i=0;i<m;i++){
23         p = (d * p + pattern[i]) % q;
24         t = (d * t + text[i]) % q;
25     }
26
27     for(int i=0;i<=n-m;i++){
28         if(p == t){
29             int j;
30             for(j=0;j<m;j++)
31                 if(text[i+j] != pattern[j])
32                     break;
33             if(j == m)
34                 cout << "Pattern found at index " << i << endl;
35         }
36         if(i < n-m){
37             t = (d*(t - text[i]*h) + text[i+m]) % q;
38             if(t < 0)
39                 t += q;
40         }
41     }
42 }
```



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6. Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\3rd Year\Semester-5\DAA_23BCS11196_KRG-2B\EXP-3.2> cd "d:\3rd Year\Semester-5\DAA_23BCS11196_KRG-2B\EXP-3.2\" ; if ($?) {
g++ Code.cpp -o Code } ; if ($?) { .\Code }
Enter the text: ababcababababd
Enter the pattern: ababd
Pattern found at index 10
```